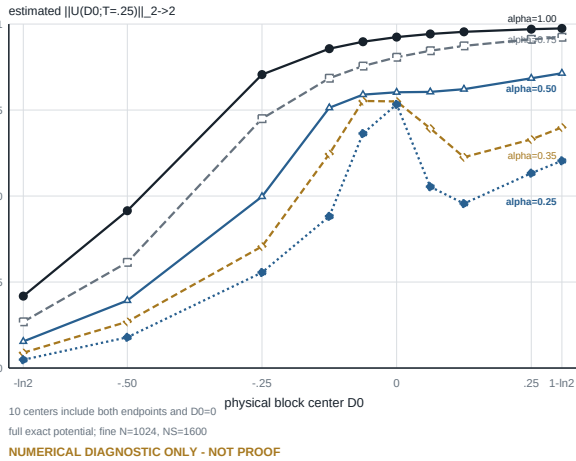
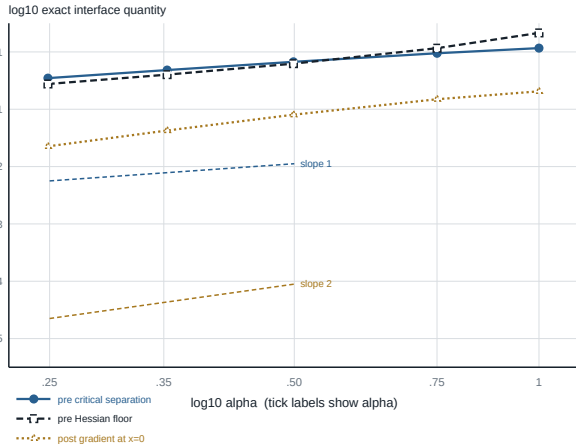


A physical-center block norm scan

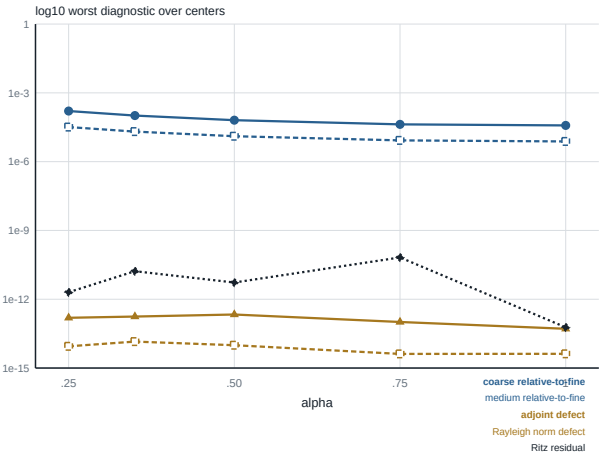


C shrinking-interface geometry



A1 fixed-shape margins vanish at $h_{\text{alpha}} = T \text{ alpha}^2$

B refinement and adjoint QA



50 center-alpha pairs x 3 refinement levels

D exact tiling and rate ledger

physical history $K^* = [-\ln 2, 1 - \ln 2]$, $T=1/4$



all-start exact tiling: $q^{\text{floor}}(L/(2T \text{ alpha}^2))$

safe envelope: $q^{\text{floor}}(L/(2T \text{ alpha}^2))$

q in (0,1) is analytic and remains nonconstructive

alpha	$h=T \text{ alpha}^2$	$N=\text{floor}(L/(2h))$	rate
1.00	0.25000	2	$\text{alpha}^{\wedge}2$
0.75	0.14062	3	$\text{alpha}^{\wedge}2$
0.50	0.06250	8	$\text{alpha}^{\wedge}2$
0.35	0.03062	16	$\text{alpha}^{\wedge}2$
0.25	0.01562	32	$\text{alpha}^{\wedge}2$

all-center scalar A2 transfer
CLOSED in bound report

forced $H^{\wedge}1$ / full linearized
nonlinear / Clay: OPEN